

Do Tendons Matter? Controllability of a Soft-PVTOL with Passive Elastic Arms

Gerardo Flores, Rodolfo Verdín, and Mark W. Spong

Abstract—We ask whether tendon actuation is necessary to fly a soft-arm aerial vehicle. In the Soft-PVTOL, a planar platform whose rigid arms are replaced by soft continuum members, tendons nominally control arm curvature, giving four inputs and five degrees of freedom. With the tendons inactive, the system retains only two thrust inputs while gaining unactuated elastic dynamics, raising a controllability question with no counterpart in rigid multirotor theory. For the physical parameter set of the prototype, we establish local controllability at hover at every admissible equilibrium, with persistence under sufficiently small parameter perturbations. The proof exploits the system’s bilateral symmetry to split the linearized dynamics into two SISO subsystems, each verified by the Popov–Belevitch–Hautus test; the enabling mechanism is the thrust-induced bending torque $T_{il} \text{sinc}(q_i)$, the sole actuation path to the arm curvatures. Simulations under asymmetric arm parameters confirm stable hover from thrust alone, while test-stand experiments with disabled tendons validate accurate pitch tracking and passive curvature convergence. Tendon-free operation is therefore a viable fallback, though at the cost of reduced control authority rather than a loss of stability.

Index Terms—Soft aerial vehicle, Soft PVTOL, Soft robotics, controllability.

I. INTRODUCTION

Morphing aerial platforms seek to increase the versatility of multirotors by adapting their geometry to the task or environment. Existing designs include foldable [1], origami-inspired [2], [3], tilt-rotor [4], and reconfigurable-arm [5], [6] vehicles. Most of these platforms, however, achieve reconfiguration through rigid joints and mechanical linkages, rather than through structurally compliant members [7]–[9].

Soft and continuously deformable arms offer an alternative way to embed morphing directly into the vehicle structure. In this direction, Flores and Spong introduced the *Soft-PVTOL* [10], [11], where the rigid arms of the classical PVTOL [12] are replaced by soft continuum arms modeled

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This letter has supplementary downloadable material provided by the authors. It contains two videos: (i) the passive-arm Soft-PVTOL test-stand validation reported in Section V, and (ii) free-flight recordings of the QuadSoft, a three-dimensional soft-arm quadrotor, operating with deactivated tendons, including an in-flight tendon-deactivation test; the latter is qualitative supporting evidence and is outside the planar analysis of this letter. It is also available at <https://github.com/gfloresc/soft-pvtol-passive-arms>.

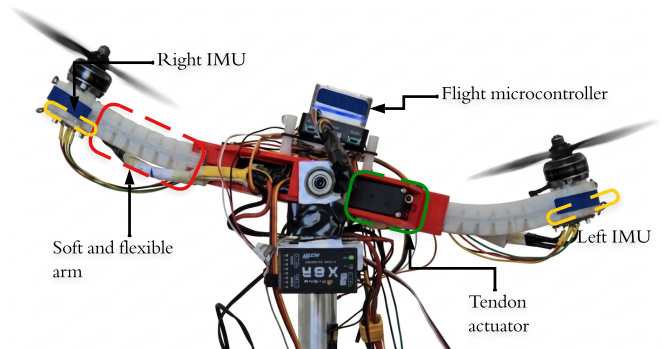


Fig. 1: Experimental Soft-PVTOL prototype on the test stand, showing the thrust-induced deflection of the flexible arms in passive mode (tendon actuators disabled). IMUs at each arm tip provide state estimation.

under the constant-curvature assumption [13]. In the nominal design, tendons actively control each arm’s curvature, giving four inputs (two thrusts and two tendon torques) for five degrees of freedom. The same concept extends to a soft quadrotor (QuadSoft), while the planar Soft-PVTOL provides a tractable analytical setting; Section V-B discusses what the planar reduction leaves out. The Soft-PVTOL prototype of Fig. 1 is used for the experimental validation reported here; free-flight recordings of the QuadSoft are included as supplementary material only, as qualitative support for tendon-free operation.

Related soft and compliant aerial vehicles have primarily used passive arm deformation for collision tolerance, impact resilience, and morphing [14]–[17]. In contrast, we ask whether the Soft-PVTOL can remain controllable when its tendon actuators are disabled: the arms then evolve as passive elastic elements, coupled to the body only through propeller thrust, and the system changes from a 5-DOF, four-input vehicle to a 5-DOF, two-input one. We show that tendon-free operation preserves controllability, albeit with reduced maneuverability, making it a viable fallback rather than the nominal mode.

Prior work on actuator loss in multirotors assumes a *rigid airframe*: the geometry is invariant under failure, and controllability reduces to the rank of a fixed allocation matrix [18]–[21]. Here the system does not simply lose inputs; it simultaneously gains unactuated, nonlinear arm dynamics coupled to the body only through the thrusts. Whether the two remaining thrust inputs can stabilize both the rigid-body motion *and* the arm deformations is a nontrivial question

with no counterpart in the rigid multirotor literature.

To our knowledge, this is the first work that addresses the controllability of a soft aerial vehicle with passive, tendon-free arms. The contributions of this paper are:

- 1) We derive the passive-arm model and show that the thrust-induced bending torque $T_i l_i \text{sinc}(q_i)$ provides the sole actuation path from the propeller thrusts to the passive arm-curvature dynamics.
- 2) For the considered passive-arm Soft-PVTOL, we establish local controllability of the system linearization about every admissible hover equilibrium, with robustness to sufficiently small parameter perturbations. The proof combines a symmetry-based decomposition into two SISO subsystems with the PBH test.
- 3) We validate the results through simulations, a stiffness-damping parametric study, and test-stand experiments.

II. THE SOFT-PVTOL MODEL

We first recall the Soft-PVTOL model from [10], then derive the passive-arm variant.

A. System Description

The Soft-PVTOL consists of a rigid body of mass m and moment of inertia I , connected to two soft continuum arms of lengths l_l and l_r with point masses m_l and m_r at their tips. Under the constant curvature assumption [13], each arm deforms into a circular arc parameterized by its total bending angle q_i , where the index $i \in \{l, r\}$ denotes the left and right arm, respectively, so that q_l and q_r are the two arm-curvature coordinates. The generalized coordinates are¹:

$$\mathbf{q} = (x_v, z_v, \theta, q_l, q_r)^\top \in \mathbb{R}^5, \quad \mathbf{u} = (T_l, T_r)^\top \in \mathbb{R}^2. \quad (1)$$

The dynamics follow the standard form [10]:

$$D(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + g(\mathbf{q}) = \tau, \quad (2)$$

where $D(\mathbf{q}) \in \mathbb{R}^{5 \times 5}$ is the symmetric, positive-definite inertia matrix, $C(\mathbf{q}, \dot{\mathbf{q}})$ is the Coriolis matrix (satisfying the skew-symmetry property $\dot{D} - 2C$), and $g(\mathbf{q})$ is the gravity vector. These matrices involve the parameter groupings $\Theta_1 = m + m_l + m_r$, $\Theta_3 = l_l m_l$, $\Theta_4 = l_r m_r$, and contain expressions such as $\sin(q_i)/q_i$ that require evaluation via their Taylor series limits as $q_i \rightarrow 0$: $\text{sinc}(q) \triangleq \frac{\sin q}{q} \rightarrow 1$, $\frac{1 - \cos q}{q} \rightarrow 0$. The complete expressions for D , C , and g are as in [10].

B. Generalized Forces and the Tendon-Free System

The thrust forces produce the first three components of the generalized force vector:

$$\begin{aligned} \tau_x &= \cos \theta (-T_r \sin q_r + T_l \sin q_l) - \sin \theta (T_r \cos q_r + T_l \cos q_l) \\ \tau_z &= \sin \theta (-T_r \sin q_r + T_l \sin q_l) + \cos \theta (T_r \cos q_r + T_l \cos q_l) \\ \tau_\theta &= l_r T_r \cos q_r \text{sinc}(q_r) - l_l T_l \cos q_l \text{sinc}(q_l), \end{aligned} \quad (3)$$

which are common to both the tendon-actuated and the tendon-free configurations. In the original tendon-actuated

Soft-PVTOL [10], two additional generalized forces, the tendon torques τ_l^t and τ_r^t , act as independent control inputs that directly command the arm curvatures, for four inputs in total. In this work we consider *the tendon-free system*, in which these tendon torques are absent and the arm curvatures are left passive. In this case the arm curvatures evolve passively under an elastic restoring torque $-k_i q_i$, viscous damping $-b_i \dot{q}_i$, and the bending moment produced by the tip thrust, whose effective moment arm under the constant-curvature model is $l_i \text{sinc}(q_i)$.

The generalized force vector becomes $\tau_{\text{passive}} = [\tau_x, \tau_z, \tau_\theta, \tau_{q_l}, \tau_{q_r}]^\top$, with $\tau_x, \tau_z, \tau_\theta$ as in (3) and the passive arm torques

$$\tau_{q_i} = -k_i q_i - b_i \dot{q}_i + T_i l_i \text{sinc}(q_i), \quad i \in \{l, r\}. \quad (4)$$

The system now has only two control inputs, $\mathbf{u} = [T_l, T_r]^\top$. The last term in each arm equation, $T_i l_i \text{sinc}(q_i)$, is the *only* mechanism through which they influence the arm curvatures in the absence of tendons; the arm coordinates are otherwise unactuated.

This raises the central question: *is the Soft-PVTOL with passive elastic arms controllable using only the thrust inputs?*

III. CONTROLLABILITY ANALYSIS

We analyze the system about hover, the nominal operating point of multirotor vehicles. For analytical clarity we first take identical arm parameters ($l_l = l_r = l$, $k_l = k_r = m_r$), which enables a decomposition into symmetric and antisymmetric modes; Remark 1 treats the mismatched case.

A. Hover Equilibrium

At hover ($\dot{\mathbf{q}} = \ddot{\mathbf{q}} = 0$, $\theta = 0$), the equilibrium condition $g(\mathbf{q}) = \tau$ yields, for the symmetric case ($l_l = l_r = l$, $k_l = k_r = k$, $m_l = m_r$):

$$T^* = \frac{g\Theta_1}{2 \cos q^*}, \quad (5)$$

$$k q^* = T^* \cdot l \cdot \text{sinc}(q^*) - g \Theta_3 \cdot G_4(q^*), \quad (6)$$

where $G_4(q) = (q \sin q + \cos q - 1)/q^2$. Several observations follow. Whenever an admissible equilibrium exists, its curvature satisfies $q^* > 0$, so the arms are never straight at hover: the thrust-induced torque deflects them into a convex shape as seen from the front, with the rotor-carrying tips raised relative to the body anchor, as visible in the passive-arm configurations of Figs. 1 and 3, and this deflection is balanced by the elastic restoring force. Consequently, from (5), the hover thrust satisfies $T^* > g\Theta_1/2$ since $\cos q^* < 1$: the deflected arms waste a portion of the thrust laterally and require more total thrust than the rigid PVTOL. Finally, as $k \rightarrow \infty$ the arms stiffen, $q^* \rightarrow 0$ and $T^* \rightarrow g\Theta_1/2$, recovering the classical PVTOL hover.

Linearizing the Euler-Lagrange dynamics (2) around the symmetric hover equilibrium

$$\mathbf{q}^* = (0, z_0, 0, q^*, q^*)^\top, \quad \mathbf{u}^* = (T^*, T^*)^\top, \quad (7)$$

¹A boldface $\mathbf{q}$ denotes the full state vector; an italic q without subscript is a generic bending angle, and scalar functions such as $\text{sinc}(q)$ act componentwise on q_l, q_r .

where $\mathbf{q}^* \in \mathbb{R}^5$ and $\mathbf{u}^* \in \mathbb{R}^2$ are the equilibrium state and input vectors, while the scalars $q^* > 0$ and $T^* > 0$ are, respectively, the common equilibrium arm curvature and the common per-rotor equilibrium thrust that satisfy the hover conditions (5)–(6) (at the symmetric equilibrium both arms share the same curvature q^* and both rotors the same thrust T^*), yields the second-order LTI model

$$M^* \ddot{\delta \mathbf{q}} + D^* \dot{\delta \mathbf{q}} + K^* \delta \mathbf{q} = B^* \delta \mathbf{u}, \quad (8)$$

with $M^* \succ 0$, and where K^* denotes the linearization of the generalized restoring forces (gravity + elasticity + thrust-induced terms) at hover.

B. Bilateral symmetry and exact decoupling

The controllability analysis exploits the bilateral symmetry of the Soft-PVTOL when the arms have identical mechanical properties. Let us define the bilateral symmetry (reflection about the vertical axis) as: $(x_v, z_v, \theta, q_l, q_r) \mapsto (-x_v, z_v, -\theta, q_r, q_l)$, and let $S \in \mathbb{R}^{5 \times 5}$ be its induced linear action at hover. Because the Lagrangian dynamics are invariant under this bilateral symmetry, the linearized matrices satisfy

$$SM^* = M^*S, \quad SD^* = D^*S, \quad SK^* = K^*S, \quad (9)$$

and the input matrix transforms by swapping left/right thrust channels:

$$SB^* = B^*P, \quad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (10)$$

Let us introduce symmetric/antisymmetric coordinates and inputs

$$\begin{aligned} q_s &= \frac{q_l + q_r}{2}, & q_a &= \frac{q_r - q_l}{2}, \\ T_s &= T_l + T_r, & T_d &= T_r - T_l, \end{aligned} \quad (11)$$

and the transformed vectors

$$\delta \tilde{\mathbf{q}} = (x_v, z_v, \theta, q_s, q_a)^\top, \quad \delta \tilde{\mathbf{u}} = (\delta T_s, \delta T_d)^\top. \quad (12)$$

There exist constant invertible matrices T_q and T_u such that $\delta \mathbf{q} = T_q \delta \tilde{\mathbf{q}}$ and $\delta \mathbf{u} = T_u \delta \tilde{\mathbf{u}}$. Using (9)–(10), the transformed second-order model becomes block diagonal:

$$\tilde{M} \ddot{\delta \tilde{\mathbf{q}}} + \tilde{D} \dot{\delta \tilde{\mathbf{q}}} + \tilde{K} \delta \tilde{\mathbf{q}} = \tilde{B} \delta \tilde{\mathbf{u}}, \quad (13)$$

where $\tilde{M} = \begin{bmatrix} M_a & 0 \\ 0 & M_s \end{bmatrix}$, $\tilde{B} = \begin{bmatrix} 0 & b_a \\ b_s & 0 \end{bmatrix}$, and the anti-symmetric block corresponds to (x_v, θ, q_a) driven only by δT_d , and the symmetric block corresponds to (z_v, q_s) driven only by δT_s . Equivalently, the system decomposes into two independent SISO subsystems:

$$\Sigma_s : M_s \ddot{\xi}_s + D_s \dot{\xi}_s + K_s \xi_s = b_s \delta T_s, \quad (14)$$

$$\Sigma_a : M_a \ddot{\xi}_a + D_a \dot{\xi}_a + K_a \xi_a = b_a \delta T_d, \quad (15)$$

where $\xi_s = (z_v, q_s)^\top \in \mathbb{R}^2$, and $\xi_a = (x_v, \theta, q_a)^\top \in \mathbb{R}^3$.

Lemma 1 (Block-diagonal controllability). *Let (A, B) be the first-order realization of (8). Under the transformation (11), (A, B) is similar to a block-diagonal pair $(\tilde{A}, \tilde{B})$ whose blocks correspond to (14) and (15). Hence, the full*

linearization is controllable if and only if both subsystems Σ_s and Σ_a are controllable.

Proof. The transformation $\delta \mathbf{q} = T_q \delta \tilde{\mathbf{q}}$, $\delta \mathbf{u} = T_u \delta \tilde{\mathbf{u}}$ induces a similarity transformation on the first-order realization of (8). Because the matrices commute with S as in (9) and the input is equivariant as in (10), the transformed matrices simultaneously preserve the symmetric/antisymmetric eigenspaces of S , yielding the block form (13). Controllability is invariant under similarity and input transformations; therefore the full pair is controllable iff each independent block is controllable. $\square$

C. Controllability of the SISO subsystems

We use the Popov–Belevitch–Hautus (PBH) test written for second-order systems. Consider

$$M \ddot{\xi} + D \dot{\xi} + K \xi = b u, \quad M \succ 0. \quad (16)$$

Let $P(\lambda) = \lambda^2 M + \lambda D + K$ be the quadratic pencil. The first-order realization of (16) is uncontrollable if and only if there exist $\lambda \in \mathbb{C}$ and $v \neq 0$ such that $P(\lambda)v = 0$, $b^\top v = 0$.

a) Input directions: At hover with $q^* \in (0, \pi/2)$, the collective and differential thrust directions are $b_s = [\cos q^*, l \operatorname{sinc}(q^*)]^\top$ and $b_a = [-\sin q^*, l \operatorname{sinc}(q^*) \cos q^*, l \operatorname{sinc}(q^*)]^\top$, whose entries are all strictly positive in magnitude whenever $q^* \in (0, \pi/2)$ (Lemma 2).

Lemma 2 (Admissible hover equilibrium). *Consider the symmetric hover equilibrium (5)–(6) and define the scalar equilibrium map $k = \kappa(q)$ obtained by substituting $T^*(q) = g\Theta_1/(2 \cos q)$ from (5) into (6),*

$$\kappa(q) = \frac{g}{2q^3} [\Theta_1 l q \tan q - 2\Theta_3 (q \sin q + \cos q - 1)], \quad (17)$$

where $q \in (0, \frac{\pi}{2})$. Then κ is continuous and positive on $(0, \pi/2)$ and satisfies $\kappa(q) \rightarrow +\infty$ as $q \rightarrow 0^+$ and as $q \rightarrow (\pi/2)^-$. Consequently κ attains a positive interior minimum $k_{\min} := \min_{q \in (0, \pi/2)} \kappa(q)$ at some $q_m \in (0, \pi/2)$, and for every $k > k_{\min}$ the equation $\kappa(q) = k$ has at least one solution on the low-curvature branch $(0, q_m)$. Any such solution satisfies $q^ \in (0, \pi/2)$, and therefore $\cos q^* > 0$, $\sin q^* > 0$, and $\operatorname{sinc}(q^*) > 0$.*

Proof. Positivity and continuity on $(0, \pi/2)$ follow from (17), since $\tan q > 0$ and, using $q \sin q + \cos q - 1 < \frac{1}{2}q^2$, the bracket is bounded below by $q^2(\Theta_1 l - \Theta_3) > 0$ because the total mass times arm length exceeds the arm first moment ($\Theta_1 l > \Theta_3$). For the boundary behaviour, as $q \rightarrow 0^+$ the bracket is $\Theta_1 l q^2 + O(q^4)$ so $\kappa(q) = \frac{g\Theta_1 l}{2q} + O(q) \rightarrow +\infty$; as $q \rightarrow (\pi/2)^-$, $\tan q \rightarrow +\infty$ gives $\kappa(q) \rightarrow +\infty$. A positive continuous function diverging at both ends of $(0, \pi/2)$ attains an interior minimum $k_{\min} = \kappa(q_m) > 0$, and by the intermediate value theorem $\kappa(q) = k$ is solvable on $(0, q_m)$ for every $k > k_{\min}$. Any such solution has $q^* \in (0, \pi/2)$, giving the stated strict positivities. Uniqueness and strict monotonicity of κ on $(0, q_m)$ hold in the physical parameter range and are verified numerically for the parameters of

Table I (a representative curve $\kappa(q)$ is monotone decreasing up to $q_m \approx 53^\circ$). $\square$

Theorem 1 (Controllability at admissible hover equilibria). *Consider the linearization (8) at an admissible hover equilibrium $q^* \in (0, \pi/2)$, which by Lemma 2 exists for every $k > k_{\min}$. For the physical parameters of Table I the PBH rank condition holds for both subsystems Σ_s and Σ_a in (14) and (15) respectively, so the linearization is controllable; by openness of controllability this persists on a neighborhood of these parameters and equilibria.*

Proof. By Lemma 1, controllability of the linearization is equivalent to controllability of the decoupled subsystems Σ_s and Σ_a . For a single-input second-order pair (M, D, K, b) , the PBH criterion states that the pair is controllable iff $\text{rank}[\lambda^2 M + \lambda D + K \mid b]$ is full for every $\lambda \in \mathbb{C}$; a rank drop at some λ requires a left null vector $w \neq 0$ with $w^\top(\lambda^2 M + \lambda D + K) = 0$ and $w^\top b = 0$.

Necessary structural condition. A rank drop requires some $w \neq 0$ orthogonal to b ; a zero entry in b supplies such a w . By Lemma 2, b_s and b_a have all entries nonzero at every admissible q^* , which removes the coordinate-aligned obstructions: this is precisely how the thrust-induced torque $T^* l \text{sinc}(q^*)$ actuates the otherwise unactuated arm coordinate.

Sufficiency. Given the nonzero input directions, the PBH rank condition for Σ_s and Σ_a is verified with the physical mass, inertia, stiffness, and damping values of Table I: evaluating the smallest singular value of $[\lambda^2 M_\bullet + \lambda D_\bullet + K_\bullet \mid b_\bullet]$ over the eigenvalues λ of each pencil and over a dense grid of admissible equilibria $q^* \in (0, \pi/2)$, this minimum remains strictly positive for both $\bullet \in \{s, a\}$, so no $\lambda \in \mathbb{C}$ makes the matrix rank-deficient. Both subsystems, and hence the full linearization, are therefore controllable. Because the pencils and input directions depend continuously on the parameters and on q^* , and controllability is an open property, this conclusion persists on a neighborhood of the physical parameters and, in particular, under sufficiently small left-right parameter mismatches, which establishes the robustness used in Remark 1. $\square$

Remark 1 (Robustness to arm-parameter mismatch). *The exact symmetric/antisymmetric decoupling relies on identical arm parameters (e.g., $k_l = k_r$, $m_l = m_r$, $l_l = l_r$). In practice, small mismatches break the exact invariance and introduce cross-coupling terms between the two modal blocks. However, controllability of LTI systems is an open property: if the nominal symmetric pair (A_0, B_0) is controllable, then there exists $\varepsilon > 0$ such that any perturbed pair (A, B) satisfying $\|A - A_0\| + \|B - B_0\| < \varepsilon$ remains controllable. Hence, the controllability result persists under sufficiently small left-right parameter variations.*

D. Dependence on the stiffness parameter

The arm stiffness k enters the linearized dynamics through the elastic potential energy, modifying both the stiffness matrix $K^*(k)$ and the equilibrium configuration $q^*(k)$; by Lemma 2 and Theorem 1, controllability then holds across

TABLE I: Physical parameters of the Soft-PVTOL, derived from the experimental prototype. Stiffness is estimated from the TPU flexural modulus and arm cross-section; damping is identified from flight data.

Parameter	Value	Parameter	Value
Body mass m	0.853 kg	Arm length l	0.110 m
Arm mass m_l, m_r	0.064 kg	Body inertia I	$3.4 \times 10^{-3} \text{ kg m}^2$
Arm inertia I_l, I_r	$5.1 \times 10^{-4} \text{ kg m}^2$	Stiffness k	2.45 N/m/rad
Damping b	0.5 N m s/rad	Hover T_{hover}	4.80 N
Sat. T_{max}	8.64 N	Total mass	0.98 kg

the whole admissible stiffness range $k > k_{\min}$, not only near a single value. The threshold $k_{\min}$ is small, below the stiffness of typical soft arms, so for the prototype and all configurations considered here $k > k_{\min}$ and the equilibrium lies well inside $(0, \pi/2)$. Below $k_{\min}$ the low-curvature equilibrium ceases to exist, which is the analytical counterpart of the low-stiffness limit of the stabilizing region in Fig. 4.

IV. CONTROL AND SIMULATION

A. Control Strategy

The simulations provide a constructive check of Theorem 1 rather than an optimized control design. With only two inputs (T_l, T_r) , we use the classical cascaded PVTOL structure, in which horizontal motion is produced indirectly through pitch. Altitude uses a PID law with feedforward gravity compensation and anti-windup, $F_z = g\Theta_1 + k_{p,z}(z_d - z_v) + k_{d,z}(\dot{z}_d - \dot{z}_v) + k_{i,z} \text{sat}(i_z)$, with a smooth C^1 quintic takeoff reference held to avoid an unrealistic instantaneous climb-rate command at $z_d = 5$ m. Horizontal position uses a PID law, whose integral term removes the small steady-state offset caused by the lateral bias of the asymmetric passive arms, $F_x = k_{p,x}(x_d - x_v) + k_{d,x}(-\dot{x}_v) + k_{i,x} \int (x_d - x_v) dt$, from which the desired pitch is $\theta_d = -\arctan(F_x / \max(F_z, F_{\min}))$, clipped to $|\theta_d| \leq 0.4$ rad. Pitch is regulated by a PD law $\tau_{\theta,d} = k_{p,\theta}(\theta_d - \theta) - k_{d,\theta} \dot{\theta}$, and the thrusts are allocated as $T_r = (F l_l + \tau_{\theta,d}) / (l_l + l_r)$, $T_l = F - T_r$ with $F = \sqrt{F_x^2 + F_z^2}$, both clipped to $[0, T_{\text{max}}]$, $T_{\text{max}} = 1.8 T_{\text{hover}}$ with $T_{\text{hover}} = g\Theta_1/2$ (Table I). The arm curvatures (q_l, q_r) are *not* controlled; they evolve as unactuated dynamics governed by (4). Gains (k_p, k_d, k_i) : (8, 4, 1.5), (1.5, 2, 0.4), (6, 2) for altitude, position, and attitude.

B. Simulation Experiments

Simulations use an RK45 integrator over $t \in [0, 16]$ s, with the prototype parameters of Table I. The initial state is $\mathbf{q}(0) = (1.0, 0.05, 0, 0.05, -0.03)^\top$, $\dot{\mathbf{q}}(0) = \mathbf{0}$, a vehicle displaced laterally with small, unequal initial curvatures. A lateral wind gust $f_{\text{wind}}(t) = 0.2 \sin(\pi(t-6)/2)$ N is injected on x_v for $6 < t < 8$ s to test robustness.

Fig. 2 shows the full state trajectories for the nominal parameter case ($k_l = k_r = 2.45$ N·m/rad, $b_l = b_r = 0.5$ N·m·s/rad, Table I). The vehicle reaches the hover set-point $(x_v^d, z_v^d) = (0, 5)$ m, the passive curvatures q_l, q_r converge to the nonzero equilibrium predicted by (5)–(6) without direct actuation, and the rotor thrusts remain smooth

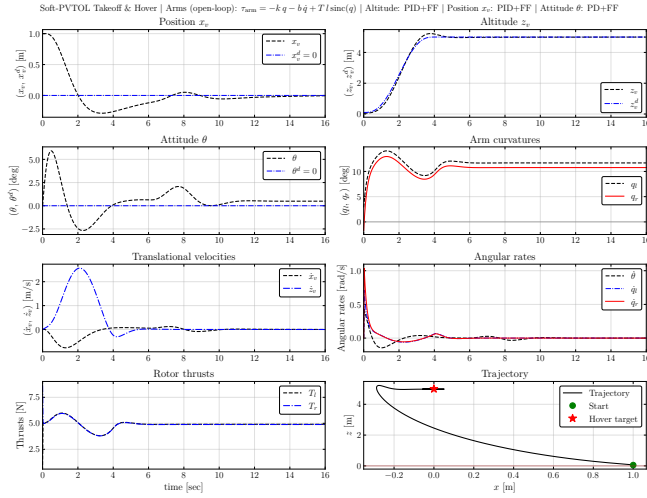


Fig. 2: State trajectories of the passive-arm Soft-PVTOL during takeoff and hover. The vehicle converges to the setpoint $(x_v^d, z_v^d) = (0, 5)$ m and the arm curvatures q_l, q_r settle passively at a nonzero equilibrium. A lateral wind gust over $t = 6$ –8 s is rejected, and the thrusts stay within the saturation limit.

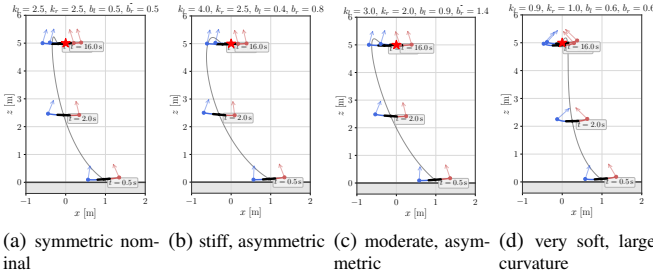


Fig. 3: Soft-PVTOL takeoff snapshots under four passive-arm parameter configurations (k and b annotated on each panel). Despite significant stiffness and damping asymmetry, the vehicle reaches the hover setpoint $(x_v^d, z_v^d) = (0, 5)$ m in all cases, showing robustness to arm parameter uncertainty.

and within the saturation limit. The lateral gust at $t = 6$ –8 s produces a bounded deviation in x_v and θ from which the system recovers, supporting the local controllability claim.

Fig. 3 presents takeoff snapshots for four asymmetric parameter configurations spanning a wide range of arm compliance. The vehicle reaches the hover setpoint in all of them, including the very-soft case of Fig. 3(d), whose equilibrium curvature exceeds 30° . Settling quality varies, since no gains were retuned per configuration, but convergence is always achieved.

To characterize the domain of validity of the result in terms of the arm parameters, Fig. 4 maps the region of the stiffness–damping plane (k, b) in which the fixed PD/PID controller drives the vehicle to the hover setpoint. The prototype operating point ($k \approx 2.45$ N/m/rad) lies comfortably inside the stabilizing region. We emphasize the distinction between this map and Theorem 1: the theorem

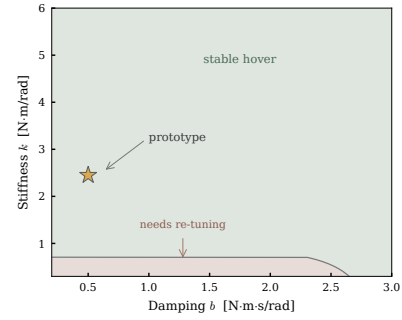


Fig. 4: Stiffness–damping (k, b) region where the PD/PID controller with fixed gains stabilizes hover (green). In the red region, the same gains no longer suffice and would require gain re-tuning. Below $k_{\min}$ no admissible equilibrium exists (Lemma 2). The prototype is marked with a star.

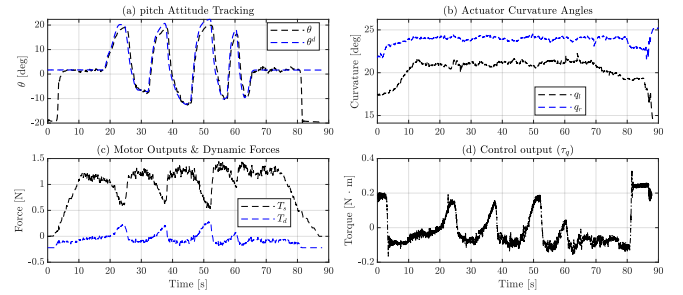


Fig. 5: Experimental validation on the test stand (translation constrained; θ, q_l, q_r free). (a) Pitch tracks θ^d via differential thrust T_d alone (no tendons). (b) Arm curvatures converge to a nonzero steady state, the q_l – q_r offset reflecting prototype asymmetry. (c)–(d) T_d modulations and torque τ_q remain bounded.

establishes *controllability* at every admissible equilibrium, whereas Fig. 4 shows the smaller region in which a *single, untuned* controller already achieves stabilization. Points inside the mapped domain but outside the stabilizing region are not uncontrollable; they simply require gain re-tuning, consistent with the theory.

V. EXPERIMENTAL RESULTS

The prototype (Fig. 1) has a 3D-printed rigid body housing a Pixhawk controller, connected to two flexible TPU arms that approximate the constant-curvature model and carry a brushless motor at each tip; IMUs at the tips measure pitch, pitch rate, and curvature. Tendon actuators are present but *disabled* here to emulate the passive-arm configuration. The vehicle is mounted on a test stand that constrains translation in x_v, z_v while leaving θ, q_l, q_r free, isolating the antisymmetric dynamics Σ_a (15). A PD controller commands differential thrust $T_d = T_r - T_l$ to track a pitch reference $\theta^d(t)$, with collective thrust held at the hover value.

A. Pitch Tracking with Passive Arms

Fig. 5(a) shows the measured pitch θ tracking a $\pm 15^\circ$ sinusoidal reference using *only* differential thrust, with tendons disabled; a minor phase lag reflects the arm compliance,

while $\dot{\theta}$ stays bounded. This confirms that the thrust-to-pitch channel of Σ_a (15) is effective on hardware. The arm curvatures (Fig. 5(b)) settle at 20° – 24° without tendon feedback, matching the nonzero equilibrium of (5)–(6), and small curvature fluctuations correlated with the thrust modulations (Fig. 5(c)) directly evidence the coupling $T_i l_i \text{sinc}(q_i)$ of (4). The q_l – q_r offset reflects the intentional stiffness asymmetry. Because the stand constrains translation, this experiment establishes accurate pitch tracking and curvature convergence with passive arms despite the asymmetry, supporting Remark 1.

B. Scope and Limitations of the Planar Analysis

The planar model isolates the thrust–elasticity coupling in a setting where it can be analyzed rigorously, at the price of effects that are structurally absent from it. Lateral bending tilts the thrust vectors out of the sagittal plane and generates yaw torques with no counterpart in the planar dynamics; aerodynamic drag on the deformed arms and residual thrust misalignment act likewise. Large passive deflection also tilts the rotors, so only a fraction $\cos q$ of each thrust acts vertically and the required total thrust grows as $1/\cos q$, consistent with (5); altitude authority therefore degrades as the arms bend further. The supplementary free-flight recordings of the QuadSoft are consistent with this picture, including a mild yaw drift on landing; they are offered as qualitative evidence that tendon-free operation is physically realizable, not as validation of the planar result, whose scope is the Soft-PVTOL. Extending the controllability analysis to the three-dimensional platform, with the yaw channel included, remains open.

VI. CONCLUSIONS

We asked whether tendon actuation is necessary to fly a soft-arm aerial vehicle, i.e. whether the Soft-PVTOL remains controllable with passive, tendon-free arms. We showed that, for the considered physical system, the passive-arm Soft-PVTOL preserves local controllability at every admissible hover equilibrium despite dropping from four inputs to two, the enabling mechanism being the thrust-induced bending torque $T_i l_i \text{sinc}(q_i)$ that couples the thrusts to the otherwise unactuated arm curvatures. Simulations confirmed that a simple PD controller drives the vehicle to a stable hover, and test-stand experiments confirmed accurate pitch tracking with passive arms, with measured deflections matching the analytical equilibrium. Tendon-free operation is therefore a viable fallback rather than the intended nominal mode, at the cost of reduced authority and not of stability. Future work includes formal stability analysis, nonlinear control for low-stiffness regimes, and extension to the three-dimensional QuadSoft.

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